



Philipp Leitner

Academic Activities (March 29, 2021)

ACQUIRED GRANTS

- Debloating AI Systems (2021), AoA SEED Grant
Co-applicant
Total budget 296,000 SEK (my part $\approx$ 128,000 SEK)
($\approx$ 13.000 EUR)
- FFI TrafCloud (2019 – 2021), VINNOVA
Co-applicant
Chalmers budget 5,500,000 SEK (my part $\approx$ 3,200,000 SEK)
($\approx$ 315.000 EUR)
- VR Starting Grant (2019 – 2022), VR
Sole applicant
Grant value 3,639,000 SEK
($\approx$ 359.000 EUR)
- WASP Academic PhD Student (2017 – 2021), WASP
Academic Supervisor
Grant value $\approx$ 3,600,000 SEK
($\approx$ 355.000 EUR)
- Chalmers AoA Hiring Startup Package (2017 – 2021), Chalmers Foundation
Sole recipient
Grant value 8,000,000 SEK
($\approx$ 789.000 EUR)
- Individual Project “MINCA” (2016 – 2019), Swiss National Science Foundation
Sole applicant
Grant value 180,000 CHF, at University of Zurich
($\approx$ 163.000 EUR)
- Technology Transfer Project “DevCloud” (2016 – 2019), Hasler Foundation
Co-applicant
Grant value 50,000 CHF, at University of Zurich
($\approx$ 45.000 EUR)

RESEARCH VISITS

Institute of of High-Performance Computing, Singapore
University of Stuttgart, Germany

Feb./Mar. 2013
Apr. 2010

INVITED TALKS AND SEMINARS

- *Unit testing performance using code microbenchmarks. How far are we?*, Invited Talk at the MongoDB Inc. Performance Seminar, November 2020
- *Performance stability of (public) clouds*, Invited Talk at Vehicle Electronics and Connected Services (VECS, industrial conference), April 2019
- *AWS Lambda and #serverless. What's all the fuzz about?*, Invited Talk at the Chaos Engineering Days Stockholm, December 2018
- *Software performance testing in a public cloud – how bad is it really?*, Invited Talk at the Metrics Day 2018, October 2018
- *Performance Testing of and in the Cloud*, Invited Lecture at the IDEA League Doctoral School on Engineering Complex Systems and IT with Big data and Information Technology, September 2018
- *Predictability of Performance in Public Clouds - Some Empirical Data and Lessons Learned for Software Performance Testing*, Seminar at IBM Research, Ireland, November 2017
- *Predictability of Performance in Public Clouds - Some Empirical Data and Lessons Learned for Software Performance Testing*, Invited Talk at the Sixth International Workshop on Load Testing and Benchmarking of Software Systems (LTB 2017), co-located with ICPE'17, April 2017
- *Your User is the Canary - Practices of and Obstacles to Conducting Experiments in Production*, CHOOSE Forum, University of Zurich, September 2016
- *How Predictable is the Performance of IaaS Instances? Some Insights from Benchmarking EC2, GCE, Azure, and Softlayer*, Open Cloud Day, ZHAW Winterthur, June 2016
- *Software Development for the Cloud – Challenges and Opportunities*, Faculty of Informatics, University of Lugano, February 2016
- *Software Development for the Cloud – Challenges and Opportunities*, VU University Amsterdam, November 2015
- *Trends, Opportunities and Challenges of Software Development for the Cloud*, Indian Institute of Technology Ropar, November 2015
- *Trends, Opportunities and Challenges of Software Development for the Cloud*, Indian Institute of Technology New Delhi, November 2015
- *Software Development for the Cloud – Challenges and Opportunities*, Chalmers, Gothenburg, June 2015
- *Engineering Java Applications for the Infrastructure-as-a-Service Cloud*, software evolution & architecture lab, University of Zurich, Zurich, October 2013
- *Building Applications for the Infrastructure-as-a-Service Cloud with CloudScale*, IMDEA Software Institute, Madrid, April 2013

- *Building Elastic Java Applications based on CloudScale and the Infrastructure-as-a-Service Paradigm*, National University of Singapore, School of Computing, March 2013
- *Building Elastic Java Applications based on CloudScale and the Infrastructure-as-a-Service Paradigm*, University of Illinois Advanced Digital Sciences Center, Singapore, February 2013

AWARDS

- Distinguished Reviewer Award (ICSME 2020)
- Distinguished Reviewer Award (MSR 2020)
- Outstanding Reviewer Award (ICPE 2020)
- Outstanding Reviewer Award (ICPE 2019)
- Best Student Paper Award (Middleware 2016)
- Best Short Paper Award (ICPC 2016)

DEPARTMENT SERVICE

- Master Program Administrator Software Engineering (Chalmers and the University of Gothenburg, from 2021)
- Member of the Faculty Senate Drafting Board (IT Faculty, University of Gothenburg, 2021 - 2024)
- Postdoc Representative at the Department of Informatics, University of Zurich (2015 - 2016)
- Member of the Habilitation Committee of Ivona Brandic (TU Vienna, 2013)
- Member of the Habilitation Committee of Karl Michael Göschka (TU Vienna, 2012)

PROFESSIONAL SERVICE

Organized Events

- Co-Organizer of the 1st and 2nd *Vienna Software Seminar (VSS)* on the Relation of Software Architecture and DevOps/Continuous Delivery. Organizers: Uwe Zdun, Philipp Leitner, Florian Rosenberg, Erik Wittern.
- Member of the Steering Committee of the International Workshop on Quality-Aware DevOps (QUDOS). From 2018.
- Dagstuhl GI Seminar on *Software Performance Engineering in the DevOps World* (September 26th – September 30th 2016, Schloss Dagstuhl, Germany). Co-organized with Andre van Hoorn, Pooyan Jamshidi, and Ingo Weber.
- Half-day *CloudWave Training Event* (September 5th 2016, TU Vienna, Austria). Half-day session at ESOC'16. Co-organized with Andreas Metzger and Eliot Salant.

Editorial Memberships

- Academic Editor of PeerJ Computer Science
- Editorial Board Member of Elsevier's Journal on Systems and Software (JSS). Responsible for coordinating and implementing the publicity efforts of the journal. 2017 - 2019.
- Special Issue Editor for "Converging Fog and Cloud Computing", in IEEE Cloud Computing. Co-edited with Erik Elmroth, Stefan Schulte, and Srikumar Venugopal.
- Special Issue Editor for "Software Tools and Technologies for Monitoring and Prediction of Cloud Services", in Software: Practice and Experience (published by Wiley-Blackwell). Co-edited with Rajiv Ranjan, Rajkumar Buyya, Armin Haller, and Stefan Tai

Chairmanships

- Program Co-Chair of 13th ACM/SPEC International Conference on Performance Engineering (ICPE)
- Awards Co-Chair of 12th ACM/SPEC International Conference on Performance Engineering (ICPE)
- Publication Chair of 10th ACM/SPEC International Conference on Performance Engineering (ICPE)
- PC Co-Chair of the 3rd International Workshop on Quality-Aware DevOps (QUDOS), co-located with the 8th ACM/SPEC International Conference on Performance Engineering (ICPE 2017)
- Co-Chair of a Special Session on Services Computing and Internet Technologies (SerCo 2016) at the International Conference on High Performance Computing & Simulation (HPCS 2016)
- Tutorial Chair of the 16th International Conference on Web Engineering (ICWE2016)
- Workshop Chair of the European Conference on Service-Oriented and Cloud Computing (ESOCC) 2015.
- General Chair of the 1st International Workshop on Monitoring and Prediction of Cloud Services (MoPoC'12), co-located with ICSOC'12.

Reviews for Funding Agencies

- Dutch Research Council (NWO)
- Natural Sciences and Engineering Research Council of Canada (NSERC)
- Deutscher Akademischer Austauschdienst (DAAD)
- Luxembourg National Research Fund (FNR)

Program Committee Memberships

(partial list)

- IEEE International Conference on Cloud Engineering (IC2E 2021)
- European Symposium on Serverless Computing and Applications (ESSCA 2020)
- International Conference on Microservices (MICROSERVICES 2020)
- IEEE/ACM International Conference on Automated Software Engineering - Demonstration Track (ASE 2019)
- IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER, from 2019)
- IEEE International Conference on Software Maintenance and Evolution (ICSME, from 2018)
- International Conference on Mining Software Repositories (MSR, from 2018)
- ACM International Conference on Distributed Event-Based Systems (DEBS 2018)
- ACM/SPEC International Conference on Performance Engineering (ICPE 2018, 2020)
- IEEE International Conference on Cloud Computing (CLOUD, 2015 – 2018)
- International Conference on the Quality of Information and Communications Technology (QUATIC, 2018)
- International Conference on Edge Computing (EDGE 2017)
- International Symposium on Fog and Edge Computing (ISFEC 2017)
- Euromicro Conference on Software Engineering and Advanced Applications (SEAA, 2018)
- International Conference on Service-Oriented Computing (ICSOC, 2015 – 2017)
- European Conference on Service-Oriented and Cloud Computing (ESOCC, 2015, 2017)
- International Conference on Web Engineering (ICWE, 2015, 2019)
- IEEE/ACM International Utility and Cloud Computing Conference (UCC, 2014 – 2017)
- International Conference on Service Oriented Computing and Applications (SOCA, 2014 – ongoing)
- Cloud Challenge, held in conjunction with UCC (2014)
- International Workshop on RESTful Designs (WS-REST, co-located with WWW, 2018)
- International Workshop on Software Engineering Aspects of Continuous Development and New Paradigms of Software Production and Deployment (DEVOPS, 2018)
- International Workshop on Autonomous Control for Performance and Reliability Trade-offs in Internet of Services (ACPROSS, 2017)
- International Workshop on Inter-Organizational Processes (IOPs, 2016)

- International Workshop on Big Data Software Engineering (BIGDSE, 2015 – ongoing)
- EAI International Conference on Cloud, Networking for IoT Systems (CN4IoT, 2015)
- International Conference on Internet and Web Applications and Services (ICIW, 2014 – 2015)
- International Conference on the Future Internet of Things and Cloud (FiCloud, 2014)
- International Conference on Service Oriented Computing (ICSOC) Demonstration Track (2014 – ongoing)
- International Workshop on Federative and Interoperable Cloud Infrastructures (FedICI, 2014)
- International Workshop on Engineering Cloud Applications & Services (EnCASE, 2014)
- Central-European Workshop on Services and their Composition (ZEUS, 2012 – ongoing)
- International Workshop on Evolutionary Business Processes EVL-BP), co-located with EDOC (2011 – 2013)
- International Workshop on Principles of Engineering Service-Oriented Systems (PESOS), co-located with ICSE (2013)
- Track on Provisioning and Management of Service Oriented Architecture and Cloud Computing (PROMASC), a Track of the WETICE (2011 – 2012)
- International Workshop on Performance Assessment and Auditing in Service Computing (PAASC), co-located with ICSOC (2011 – 2012)
- Workshop on Emerging Web Services Technology (WEWST), co-located with ECOWS (2009 – 2011)
- International Workshop on Dynamic and Declarative Business Processes (DDBP), co-located with EDOC (2010)

Reviews for Journals

(partial list)

- Elsevier Information and Software Technology (IST)
- IEEE Transactions on Software Engineering (TSE)
- Springer Empirical Software Engineering (ESEM)
- ACM Transactions on Internet Technology (TOIT)
- IEEE Transactions on Cloud Computing (TCC)
- Wiley Concurrency and Computation: Practice and Experience
- Oxford Press: The Computer Journal
- International Journal of Communication Networks and Distributed Systems (IJCNDS)
- Elsevier Information Processing Letters (IPL)

- IEEE Internet Computing (IC)
- IEEE Software
- Elsevier Journal of Parallel and Distributed Computing (JPDC)
- ACM Transactions on the Web (TWEB)
- Elsevier Journal of Systems and Software (JSS)
- Springer Computing (Archives for Scientific Computing)
- Journal of Computer Science and Technology (JCST)
- International Journal of Cooperative Information Systems (IJCIS)
- IEEE Transactions on Services Computing (TSC)
- Transactions on Pattern Languages of Programming (TPLoP)

RESEARCH STUDENTS

Current PhD Students

- **Hazem (Peter) Samoaa** (ML for performance prediction)
- **Hamdy Michael Ayas** (architectural decision making in cloud computing, *co-supervisor*)
- **Linda Erlenhov** (bots in software development)
- **Joel Scheuner** (cloud benchmarking)

Graduated PhD Students

- **Christoph Laaber** (UZH, microbenchmarking in CI builds, co-supervised with Harald C. Gall, now postdoc at Simula (Norway))
- **Gerald Schermann** (UHZ, continuous experimentation, co-supervised with Harald C. Gall, now release engineer at digitec.ch)
- **Jürgen Cito** (UZH, cloud performance engineering, co-supervised with Harald C. Gall, now Assistant Professor at TU Wien)

PhD Committee Appointments

- **Luca Traini** (Università degli Studi dell'Aquila, assessor)
- **Hugo Sica de Andrade** (Chalmers University of Technology, substitute member of grading committee)
- **Lars Larsson** (Umeå University, faculty opponent)
- **Alfred Åkesson** (Lund University, licenciate discussion leader)